1) a)

$$\left(\mathbf{r}-\begin{pmatrix}2\\8\\1\end{pmatrix}\right)\times\begin{pmatrix}5\\-2\\3\end{pmatrix}=\mathbf{0}$$

1) b)

$$\left(\mathbf{r}-\begin{pmatrix}-10\\3\\-5\end{pmatrix}\right)\times\begin{pmatrix}7\\0\\-1\end{pmatrix}=\mathbf{0}$$

2)

$$\left(\begin{pmatrix}2\\0\\3\end{pmatrix}-\begin{pmatrix}6\\3\\-2\end{pmatrix}\right)\times\begin{pmatrix}4\\3\\-5\end{pmatrix}=\begin{pmatrix}-4\\-3\\5\end{pmatrix}\times\begin{pmatrix}4\\3\\-5\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-4&-3&5\\4&3&-5\end{vmatrix}$$

$$=\big((-3)\times(-5)-5\times3\big)\mathbf{i}-\big((-4)\times(-5)-5\times4\big)\mathbf{j}+\big((-4)\times3-(-3)\times4\big)\mathbf{k}$$

$$=0\mathbf{i}+0\mathbf{j}+0\mathbf{k}=\mathbf{0}$$

$\therefore (2,0,3)$ lies on the line

3) a)

$$\begin{pmatrix}5\\4\\-2\end{pmatrix}-\begin{pmatrix}-3\\0\\9\end{pmatrix}=\begin{pmatrix}8\\4\\-11\end{pmatrix}$$

$$\therefore\left(\mathbf{r}-\begin{pmatrix}-3\\0\\9\end{pmatrix}\right)\times\begin{pmatrix}8\\4\\-11\end{pmatrix}=\mathbf{0}$$

3) b)

$$\begin{pmatrix}-3\\9\\25\end{pmatrix}-\begin{pmatrix}-2\\-1\\-6\end{pmatrix}=\begin{pmatrix}-1\\10\\31\end{pmatrix}$$

$$\therefore\left(\mathbf{r}-\begin{pmatrix}-2\\-1\\-6\end{pmatrix}\right)\times\begin{pmatrix}-1\\10\\31\end{pmatrix}=\mathbf{0}$$

4)

$\begin{pmatrix}4\\2\\1\end{pmatrix} = k\begin{pmatrix}2\\1\\0\end{pmatrix}$

$4 = k(2) \Rightarrow k = 2$

$2 = k(1) \Rightarrow k = 2$

$1 = k(0) \Rightarrow k = 0$

Inconsistent values of k, so the lines are not parallel (or coincident).

5) a)

$l_1: \mathbf{r} = \begin{pmatrix}3\\-8\\4\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\4\end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix}6\\5\\-3\end{pmatrix} + \mu\begin{pmatrix}-4\\3\\1\end{pmatrix}$

$2 = k(-4) \Rightarrow k = -\frac{1}{2}$

$1 = k(3) \Rightarrow k = \frac{1}{3}$

$4 = k(1) \Rightarrow k = \frac{1}{4}$

Inconsistent values of k, so the lines are not parallel (or coincident).

$3 + 2\lambda = 6 - 4\mu \Rightarrow 2\lambda + 4\mu = 3$ (1)

$-8 + \lambda = 5 + 3\mu \Rightarrow \lambda - 3\mu = 13$ (2)

$4 + 4\lambda = -3 + \mu \Rightarrow 4\lambda - \mu = -7$ (3)

Solve (1) and (2): $\lambda = \frac{61}{10}$ and $\mu = -\frac{23}{10}$

Check in (3):

$4\left(\frac{61}{10}\right) - \left(-\frac{23}{10}\right) = \frac{267}{10} \neq -7$

So the lines do not intersect.

l_1 and l_2 are skew lines

5) b)

$l_1: \mathbf{r} = \begin{pmatrix}5\\4\\2\end{pmatrix} + \lambda\begin{pmatrix}-1\\-5\\1\end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix}3\\3\\4\end{pmatrix} + \mu\begin{pmatrix}6\\5\\-6\end{pmatrix}$

$-1 = k(6) \Rightarrow k = -\frac{1}{6}$

$-5 = k(5) \Rightarrow k = -1$

$1 = k(-6) \Rightarrow k = -\frac{1}{6}$

Inconsistent values of k, so the lines are not parallel (or coincident).

$5 - \lambda = 3 + 6\mu \Rightarrow -\lambda - 6\mu = -2$ (1)

$4 - 5\lambda = 3 + 5\mu \Rightarrow -5\lambda - 5\mu = -1$ (2)

$2 + \lambda = 4 - 6\mu \Rightarrow \lambda + 6\mu = 2$ (3)

Solve (1) and (2): $\lambda = -\frac{4}{25}$ and $\mu = \frac{9}{25}$

Check in (3):

$\left(-\frac{4}{25}\right) + 6\left(\frac{9}{25}\right) = 2$ as required

So the lines intersect.

Substitute $\lambda = -\frac{4}{25}$ into l_1:

The lines intersect at $\left(\frac{129}{25}, \frac{24}{5}, \frac{46}{25}\right)$

6)

$\cos\theta = \frac{\begin{pmatrix}2\\-9\\0\end{pmatrix}\cdot\begin{pmatrix}7\\6\\1\end{pmatrix}}{\left\|\begin{pmatrix}2\\-9\\0\end{pmatrix}\right\|\left\|\begin{pmatrix}7\\6\\1\end{pmatrix}\right\|} = \frac{2\times7+(-9)\times6+0\times1}{\sqrt{2^2+(-9)^2+0^2}\sqrt{7^2+6^2+1^2}} = \frac{-40}{\sqrt{85}\sqrt{86}} = -\frac{40}{\sqrt{7310}}$

$\theta = \arccos\left(-\frac{40}{\sqrt{7310}}\right) = 117.894\ldots°$

Acute angle is $180° - 117.894\ldots° = 62.1°$ (1 DP)

7) a)

$l_1: \mathbf{r} = \begin{pmatrix} 5 \\ -8 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -7 \\ 1 \end{pmatrix}$

$l_2: \mathbf{r} = \begin{pmatrix} -9 \\ 10 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -13 \\ 1 \\ -1 \end{pmatrix}$

$5 + 3\lambda = -9 - 13\mu \Rightarrow 3\lambda + 13\mu = -14$ (1)

$-8 - 7\lambda = 10 + \mu \Rightarrow -7\lambda - \mu = 18$ (2)

$6 + \lambda = 3 - \mu \Rightarrow \lambda + \mu = -3$ (3)

Solve (1) and (2): $\lambda = -\frac{5}{2}$ and $\mu = -\frac{1}{2}$

Check in (3):

$\left(-\frac{5}{2}\right) + \left(-\frac{1}{2}\right) = -3$ as required

So the lines intersect.

Substitute $\lambda = -\frac{5}{2}$ into l_1:

The lines intersect at $\left(-\frac{5}{2}, \frac{19}{2}, \frac{7}{2}\right)$

7) b)

$\cos\theta = \frac{\begin{pmatrix} 3 \\ -7 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -13 \\ 1 \\ -1 \end{pmatrix}}{\left|\begin{pmatrix} 3 \\ -7 \\ 1 \end{pmatrix}\right| \left|\begin{pmatrix} -13 \\ 1 \\ -1 \end{pmatrix}\right|} = \frac{3\times(-13)+(-7)\times1+1\times(-1)}{\sqrt{3^2+(-7)^2+1^2}\sqrt{(-13)^2+1^2+(-1)^2}}$

$= \frac{-47}{\sqrt{59}\sqrt{171}} = -\frac{47}{\sqrt{10089}}$

$\theta = \arccos\left(-\frac{47}{\sqrt{10089}}\right) = 117.899\ldots°$

Acute angle is $180° - 117.899\ldots° = 62.1°$ (1 DP)

7) c)

$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 0 \\ 15 \end{pmatrix}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 5+3\lambda \\ -8-7\lambda \\ 6+\lambda \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 15 \end{pmatrix} = \begin{pmatrix} 4+3\lambda \\ -8-7\lambda \\ -9+\lambda \end{pmatrix}$

$\overrightarrow{AB}$ is perpendicular to l_1 so:

$\begin{pmatrix} 3 \\ -7 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4+3\lambda \\ -8-7\lambda \\ -9+\lambda \end{pmatrix} = 0$

$\Rightarrow 3 \times (4+3\lambda) - 7 \times (-8-7\lambda) + 1 \times (-9+\lambda) = 0$

$\Rightarrow 12 + 9\lambda + 56 + 49\lambda - 9 + \lambda = 0$

$\Rightarrow 59\lambda + 59 = 0$

$\Rightarrow \lambda = -1$

$\overrightarrow{OB} = \begin{pmatrix} 5+3(-1) \\ -8-7(-1) \\ 6+(-1) \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$

So B has coordinates $(2, -1, 5)$

8) a)

$\mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 9 \\ 1 \end{pmatrix}$

8) b)

$\mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 2 \\ -3 \end{pmatrix}$

9) a)

$\frac{x-3}{4} = \frac{y-2}{-2} = \frac{z+9}{1}$ or $\frac{x-3}{4} = \frac{2-y}{2} = z + 9$

9) b)

$\frac{x-8}{-2} = \frac{y+1}{-3} = \frac{z-0}{1}$ or $\frac{8-x}{2} = \frac{y+1}{-3} = z$

9) c)

$\frac{x-0}{1} = \frac{z-9}{-2}$, $y = 2$ or $x = \frac{9-z}{2}$, $y = 2$

10) a)

Vector Form:

$$\mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}$$

Cartesian Form:

$x = 3 + \lambda \Rightarrow \lambda = x - 3$

$y = -2 + 4\lambda \Rightarrow \lambda = \frac{y+2}{4}$

$z = 4 + 2\lambda \Rightarrow \lambda = \frac{z-4}{2}$

$\therefore x - 3 = \frac{y+2}{4} = \frac{z-4}{2}$

10) b)

Vector Form:

$$\mathbf{r} = \begin{pmatrix} 3 \\ -7 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 4 \\ -5 \end{pmatrix}$$

Cartesian Form:

$x = 3$

$y = -7 + 4\lambda \Rightarrow \lambda = \frac{y+2}{4}$

$z = 0 - 5\lambda \Rightarrow \lambda = \frac{z}{-5}$

$\therefore x = 3, \frac{y+2}{4} = \frac{z}{-5}$